EXAMPLE DATA - generated in demo/mock mode. Areas of interest, findings and work orders are illustrative and NOT field-verified results.

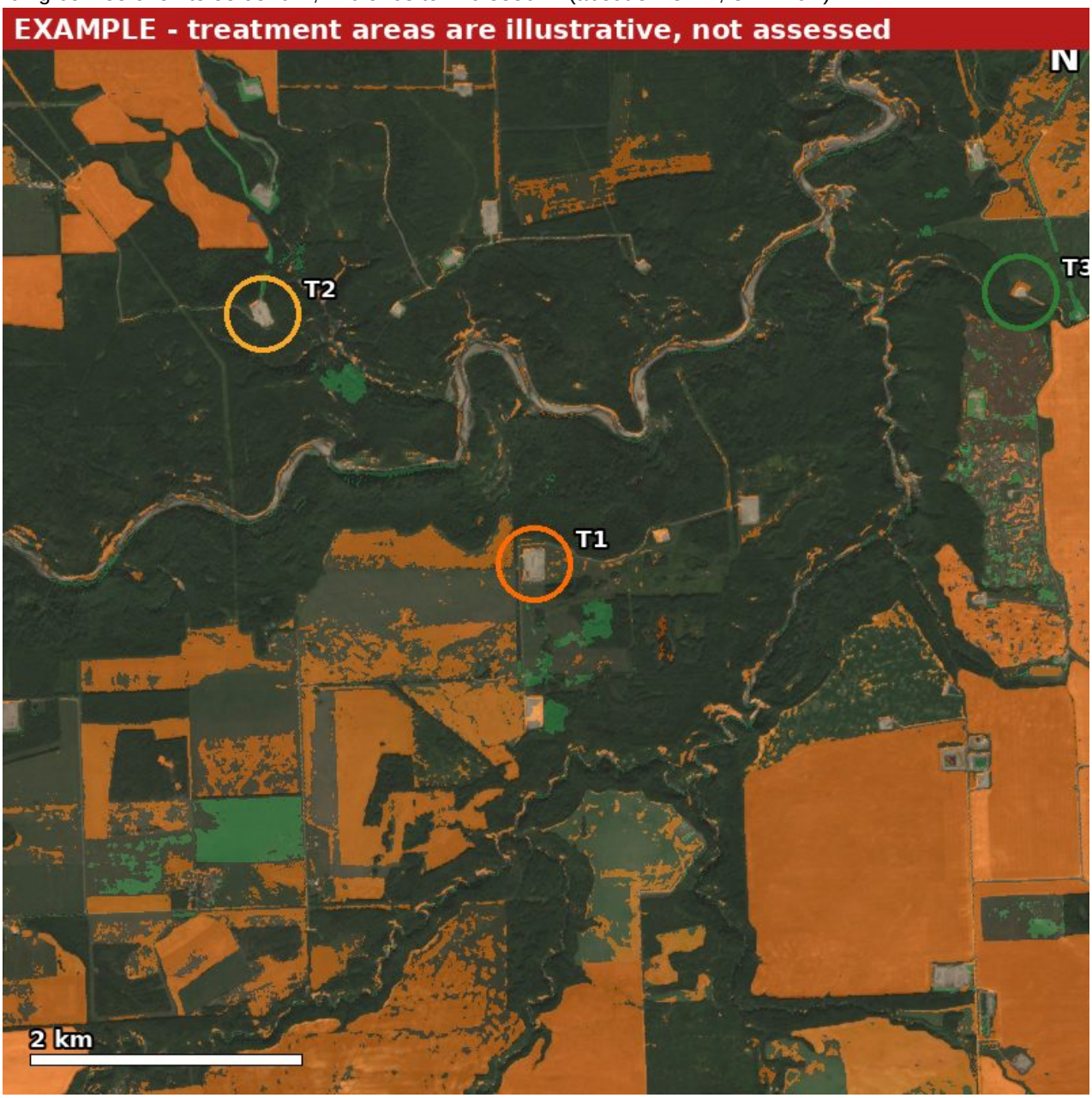
Community Wildfire Mapping Report

EXAMPLE Community - Peace Region study area | Prepared 2026-10-16 | Centre 55.9500 N, 120.4500 W

Supporting wildfire mapping for a UBCM Community Resiliency Investment (CRI) FireSmart application or a Community Wildfire Resiliency Plan. Check the current program guide for required content; this report does not replace a qualified professional's assessment.

1. Area of interest

Bounding box: 55.9154 to 55.9846 N, 120.5163 to 120.3836 W (about 8 x 8 km, UTM 10N).



Map: Sentinel-2 true colour (2021) blended with the NDVI change layer (orange = NDVI drop > 0.15, green = gain > 0.15); circles mark treatment areas T1..Tn. Contains modified Copernicus Sentinel data [2018, 2021].

2. Vegetation change (NDVI)

Item	Value
Before scene	2018-08-02 (S2A_10VFH_20180802)
After scene	2021-09-02 (S2A_10VFH_20210902)
Pixel size	10 m
Share of AOI with NDVI drop > 0.15	25.5% (computed from the scenes)

Most of the drop in this scene pair is farm harvest and the seasonal difference (early August vs early September), not wildfire or fuel change. NDVI change flags where to look; it does not measure fuel load.

3. Wildfire risk rating (historical exposure)

Measure	Value
Historical exposure rating	**MODERATE** (score 48.2 / 100)
Fires within 10km	5 fires, 98 ha
Fires within 25km	37 fires, 1,650 ha
Fires within 50km	87 fires, 10,296 ha
Cause mix within 25 km	human 73%, lightning 0%, unknown 27% (37 fires)
Nearest fire >= 1,000 ha	Boundary Lake (2023), 6,422 ha, 48.0 km

Computed by SiteSense from BC Wildfire Service historical fire incident locations (2015-2025). Historical 2015-2025 exposure, not a forecast. Check current BC Wildfire Service danger ratings and bans.

4. Prioritized treatment areas

EXAMPLE: these areas were hand-placed on visible features of the imagery to illustrate the format. They are not an assessment of fuel hazard.

#	Priority	Location	Basis	Suggested next step
T1	High	55.9492, -120.4514 exposure: moderate	Cleared pad near a linear corridor; example of where to check pad-edge vegetation for stress	Field-check fuel loading at the pad edge; candidate for fuel thinning / surface fuel removal.
T2	Medium	55.9663, -120.4823 exposure: moderate	Bright cleared pad (likely wellsite/lease) with adjacent darker patch	Check clearance between the cleared pad and adjacent forest edge; candidate for a fuel break.
T3	Low	55.9659, -120.3929 exposure: moderate	Small bright clearing; possible new disturbance since 2018	Confirm cause of disturbance; check for slash or debris piles that add fuel.

Ranking: imagery priority first, then historical wildfire exposure at each point. Field verification by a qualified professional is required before any treatment.

5. Methods

1) Sentinel-2 L2A scenes retrieved from the Element84 Earth Search STAC (AWS Open Data); NDVI = (B08 - B04) / (B08 + B04) computed for each date and differenced. 2) Candidate areas flagged on the change map (hand-placed in this

EXAMPLE). 3) Historical exposure: BCWS fire origins (>= 1 ha) counted within 10, 25 and 50 km and scored 0-100 (weights calibrated against Emberline town scores; rating bands watch to extreme). 4) Areas ranked by imagery priority, then exposure.

6. Data sources and licences

Data	Source	Licence / attribution
Historical fire incident locations 2015-2025	BC Wildfire Service (BC Data Catalogue)	Open Government Licence - British Columbia
Sentinel-2 L2A imagery	Copernicus / ESA via Element84 Earth Search STAC	Contains modified Copernicus Sentinel data [2018, 2021].
Risk scoring method	SiteSense / Emberline	Historical exposure, not a forecast

Contains information licensed under the Open Government Licence - British Columbia.

7. Limitations

Fire data are origin points, not perimeters. 10 m imagery cannot resolve individual trees or surface fuels. Ratings reflect 2015-2025 history and are not a fire danger forecast; check current BC Wildfire Service danger ratings, bans and evacuation orders.